SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1714**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I S.Praveen (192524239) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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20 + 20 + 17 + 17 + 8

82

1) Vehicle Fault Diagnosis system:

a) Propositional Logic Representation:

- E – Engine warning Light
- N – Unusual Noise
- O – Low oil pressure
- H – Overheating sign
- F – Possible Engine Failure Risk
- L – Possible Oil Leak
- C – Possible Coolant System failure.

Rules:

1. $E \wedge N \rightarrow F$ $\quad E = True$
2. $E \wedge O \rightarrow L$ $\quad N = True$
3. $N \wedge H \rightarrow C$ $\quad H = True$

So, initial knowledge base

$KB = \{E, N, H, (E \wedge N \rightarrow F), (E \wedge O \rightarrow L), (N \wedge H \rightarrow C)\}$

b) Forward chaining:

Initial Facts :- E, N, F

Step1 – Apply Rule 1

Rule 1 : $E \wedge N \rightarrow F$

Since : $E \rightarrow True$, $N \rightarrow True$

both conditions are satisfied.

Therefore, $\boxed{F = True}$

Step2 : Rule 2 :

$E \wedge O \rightarrow L$ We know

$E = True$

②

but Low Oil Pressure (O) is not given.

$\boxed{\text{L cannot be derived}}$

Final

No conclusion about possible oil leak can be made because Low Oil pressure is unavailable.

c) Backward chaining

Goal = Coolant system failure (x)

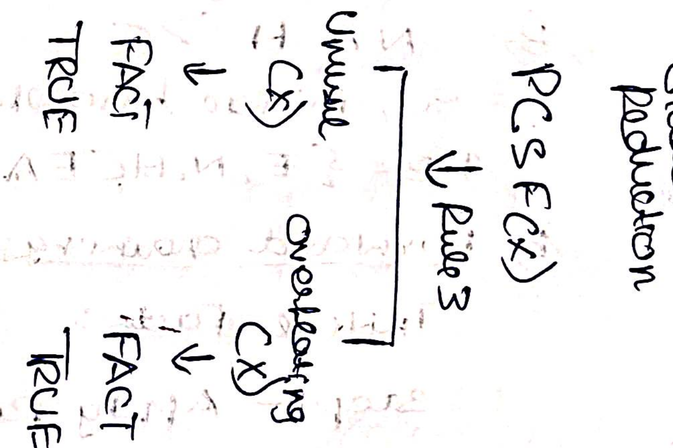
Possible coolant system failure (x)

Goal: Possible coolant System Failure(x)

UnusualNoise(x) ∧ overheating sign(x) →

Possible coolant system failure(x)

$\boxed{\text{∴ Possible coolant system Failure(x) is proved.}}$



2) Hospital Emergency Ward Routing Agent:-

a) Unification of Rule 1

Rule 1:-

$\forall x$ Patient (x) ∧ Critical Condition (x) →

Priority Corridor(x)

(3)

Patient (Patient A)

Critical condition (Patient A)

Patient(x)

Patient (Patient A)

x/Patient A precises,

$\theta = \{x \rightarrow \text{patient A}\}$

Critical condition(x)

Critical condition (patien A)

Priority Corridor (Patient A)

$\theta = \{x/\text{Patient A}\}$

b) **Resolution Proof on Hold:**

Rule1: Patient(x) $\wedge$ Critical condition (x) $\rightarrow$ Priority Corridor

$\neg$(Patien(x) $\wedge$ Critical condition(x)) $\vee$ Priority Corridor(x)

De-Morgan law:

$\neg$ Patient (x) $\vee$ $\neg$ Critical condition(x) $\vee$ Priority corridor(x)

Resolution clain

Patient(A) +

Critical condition(A)

$\Leftarrow$

Priority Corridor(A)

$\Leftarrow$

Corridor cleared(A)

Patient(B) + Waiting Board (B, A)

$\Leftarrow$

Hold(B)

c) Patient(PatientC)
Critical condition (Patient c)
Priority carridor
carridor closed (Patient c)

∴ Waiting Behind (PatientC, Patient A)

3) Weather Based flight schoduling Expert system

Rules;
$P1 : S \to D$
$P2 : D \wedge I \to N$
$P3 : \neg N \wedge I \to C$

Facts; S= True, I = true

a) P1 :- $S \to D$
$\neg S \vee D$
$P1 = (\neg S \vee D)$

P2: $D \wedge I \to N$
$\neg(D \wedge I) \vee N$
$(\neg D \vee \neg I) \vee N$
$\boxed{\neg D \vee \neg I \vee N}$

P3: $\neg N \wedge I \to C$
$\neg(\neg N \wedge I) \vee C$
$(N \vee \neg I) \vee C$

$N \vee \neg I \vee C$

$\neg S \vee D$
$\neg D \vee \neg I N$
$N \vee \neg I \vee C$

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b) $\neg S \vee D$
$\neg D \vee \neg I \vee N$
$\neg I \vee N$

$\neg S \vee D$
$\neg D \vee \neg I \vee N$
$\neg I \vee N$
S
$\square$

NotifyPassengers is logally proved

c) S = True, I = true
$S \Rightarrow D$
$\neg S \vee D$

FlightDelay Needed cannot be derived

$D \wedge I$
$\neg N \wedge I \rightarrow C$
Compliance Risk cannot be detected.

4) a) B_0 = {Draft[1,3], Glow[1,1], Sparkle[2,3]}
Step1 – Draft
Draft[1,3] → PitAdjacent [1,3]
Draft[1,3]
PitAdjacent [1,3]
x = 2, y = 3
Treasure [2,3]
PitAdjacent [1,3]
Lament Adjacent[1,1]
Treasure [2,3]

⑥

b) Draft [1, 3]

Pit Adjacent [1, 3]

Pit adjacent [1, 3]

Crow [1, 1] → Lava Adjacen [1, 1]

Sparkle [2, 3]

Sparkle [x, y] → Treasure [x, y]

Treasure [2, 3]

c) [1, 1] → [1, 3] → [2, 3]

Lava Vent Adjacent [1, 1]

Pit adjacent [1, 3]

Treasure [2, 3]

$\neg$Pit [x, y] $\wedge$ $\neg$ Lava Vent [x, y]

Safe [x, y]

$\neg$Pit [1, 3] , $\neg$ Lava vent [1, 3]

So, $\neg$Pit [1, 1] $\wedge$ $\neg$ Lava Vent [1, 1]

$\neg$Pit [1, 3] $\wedge$ $\neg$ Lava Vent [1, 3]

$\neg$Pit [2, 3] $\wedge$ $\neg$ Lava Vent [2, 3]